

Machine Learning Classification Model Comparison Report

1. Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9444	0.8919	0.9444	0.9174
Decision Tree	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000
Gradient Boosting	1.0000	1.0000	1.0000	1.0000

2. Conclusion

Based on the evaluation, the Decision Tree, Random Forest, and Gradient Boosting models achieved perfect accuracy (100%) on the test set. Logistic Regression performed well with 94.4% accuracy but struggled to identify the minority class. The tree-based models are recommended for this dataset as they effectively capture the non-linear patterns present in the anomaly detection data.